Linguistics Knowledge and NLP Challenges

Dr. Saeedeh Momtazi

دانشگاه صنعتی امیرکبیر
(پلی تکنیک تهران)

Outline

- **Linguistic Knowledge**
- Challenges

Linguistic

- Phonetics and Phonology
- Morphology
- Syntax
- Semantics
- Pragmatics
- Discourse

Phonetics and Phonology

- The study of linguistic sounds and their relations to words

b	b	l	l	sh	ʃ	a	æ	oh	ɔː^{r}
p	p	r	ɹ	zh	ʒ	ah	aː^{r}	oa	o
d	d	m	m	th	θ	ay	e	u	ʊ
t	t	n	n	h	h	e	ɛ	uh	ʌ
f	f	s	s	w	w	ee	i	oo	u
v	v	z	z	y	j	i	ɪ	oi	ɔj
g	ɡ	ch	tʃ	ng	ŋ	iy	aj	ow	aw
k	k	j	dʒ			o	ɑ		

Morphology

- The study of internal structures of words and how they can be modified
- Parsing complex words into their components

unbelievable

un+believ+able

prefix: un

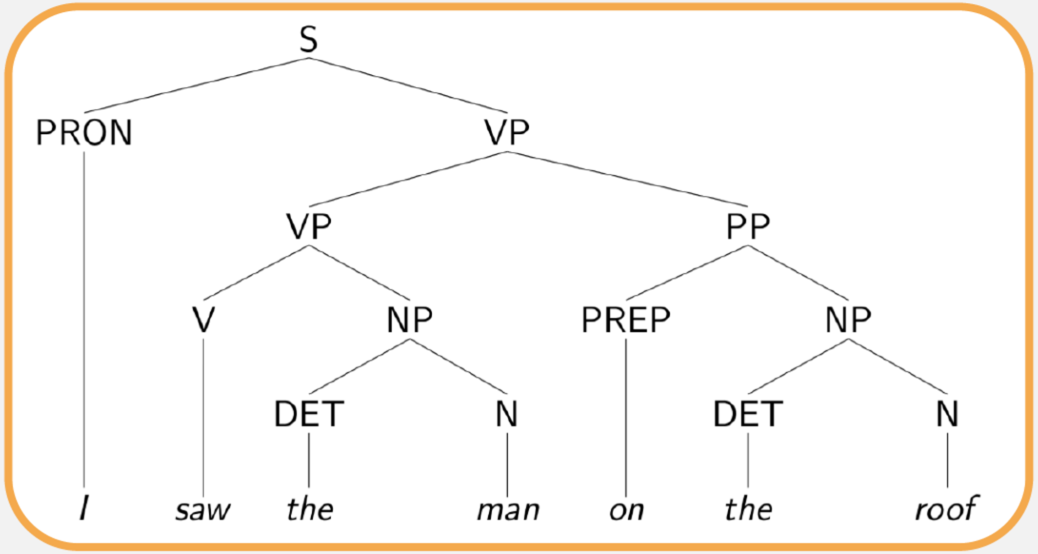
root: believe

sufix: able

Syntax

The study of the structural relationships between words in a sentence

For Example: I saw the man on the roof.



Semantics

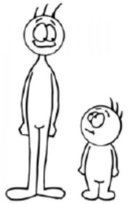
- The study of the meaning of words, and how these combine to form the meanings of sentences
- Realizing lexical relations among words

Hyponymy (is a)

Synonymy

Meronymy (part of)

Antonymy



Pragmatics

- The study of how language is used to accomplish goals, and the influence of context on the meaning
- Understanding the aspects of a language which depends on situation and world knowledge

? Do you know what time it is?

Discourse

- The study of linguistic units larger than a single utterance

For Example:

John reads a book. He borrowed it from his friend.

Outline

- Linguistic Knowledge
- **Challenges**

Challenges

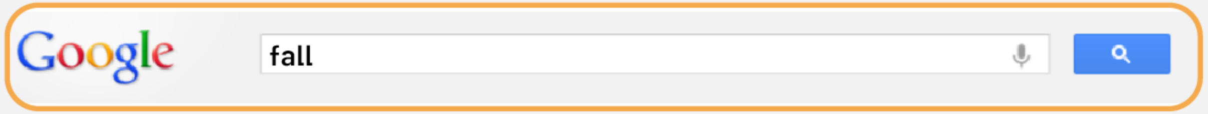
- Different words/sentences express the same meaning
  - The third season of the year
    - Fall
    - Autumn
  - Book delivery time
    - When will my book arrive?
    - When will I receive my book?

Paraphrasing

- One word/sentence can have different meanings
  - Fall
    - The third season of the year
    - Moving down towards the ground or towards a lower position
  - The door is open.
    - Expressing a fact
    - A request to close the door

Ambiguity

Challenges



Challenges

What about autumn?

Phonetics and Phonology

Computers can recognize speech

Computers can wreck a nice peach

Unionize

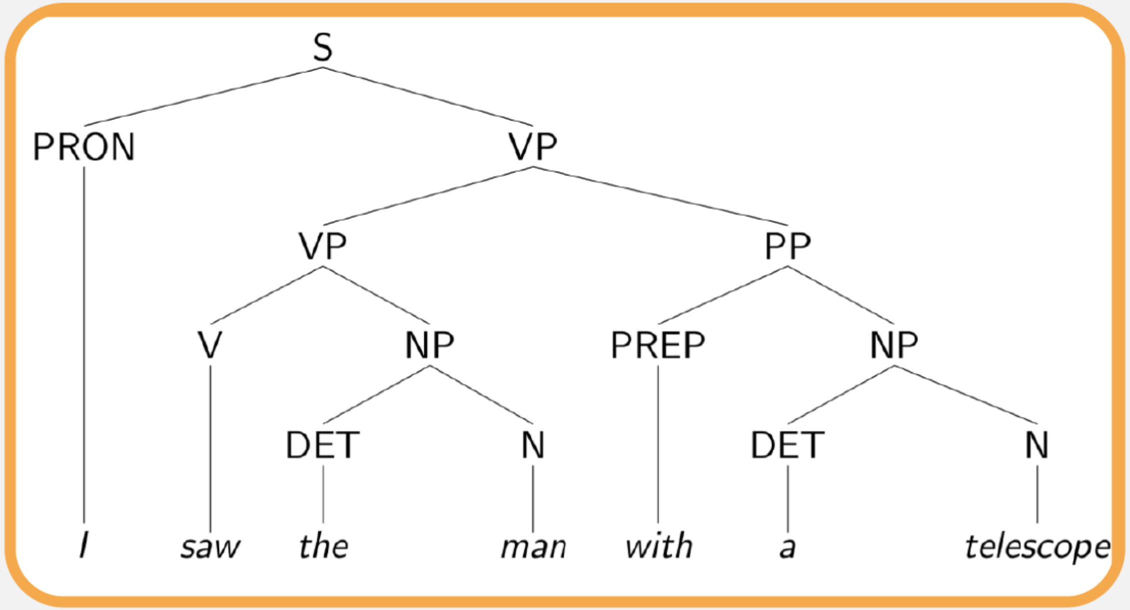
Union + ize

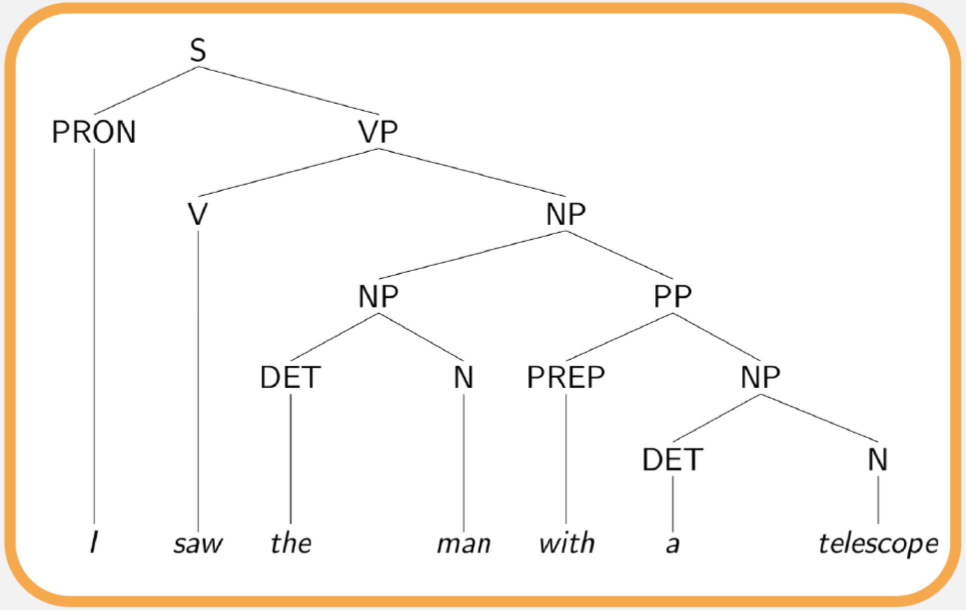
Un + ion + ize

Syntax

For Example:

I saw the man with a telescope.





Semantics

The astronomer loves the star

- Movie Star (Celebrity)
- Sky Star

Semantics

Every man loves a woman.

$$\forall x : (man(x) \longrightarrow \exists y : (woman(y) \ \& \ love(x, y)))$$

$$\exists y : (woman(y) \ \& \ \forall x : (man(x) \longrightarrow love(x, y)))$$

Pragmatics

Can you give me the salt?

- Would you please give me the salt?
- Do you have the ability to give me the salt?

Discourse

Alice understands that you like your mother, but she ...

Alice

Your Mother

Ambiguity

I made her duck.

Ambiguity

I made her duck.

- I cooked duck for her (to eat)
- I cooked duck belonging to her
- I created a toy duck which she owns
- I created a vehicle which she owns
- I caused her to quickly lower her head or body
- I waved my magic wand and turned her into a duck

Ambiguity

- duck - morphologically and syntactically ambiguous
  - noun
  - verb
- duck - semantically ambiguous
  - waterfowl
  - vehicle moving on the water
- her - syntactically ambiguous
  - possessive
  - dative
- make - semantically ambiguous
  - cook
  - create
- make - syntactically ambiguous
  - Takes a direct object and a verb (5)
  - Transitive - takes a direct object (2)
  - Di-transitive - takes two objects (6)

I made her duck